Uncertainty-Aware SegFormer for Satellite Image Segmentation Using Evidential Deep Learning

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***Abstract*—Vision Transformers and other deep learning models are highly accurate in segmenting satellite images, but they tend to behave as overconfident black boxes, and pose a safety risk for critical remote sensing applications. We present an Uncertainty-Aware SegFormer that combines Evidential Deep Learning (EDL) with a single forward pass to predict epistemic uncertainty. On the DeepGlobe Land Cover dataset, we reach a mean Intersection over Union (mIoU) of 65.81%. Through a series of experiments, we vary the weight of the Kullback-Leibler (KL) regularization, and find a trade-off: a larger regularization weight increases the correlation between uncertainty and error, but it also decreases spatial accuracy and calibration performance. There is no optimal setting for all three objectives. Compared to Monte Carlo (MC) Dropout, our single-pass EDL method provides better uncertainty-error localization, though MC Dropout achieves slightly better accuracy and calibration at the cost of 1.5 times more computing time. Furthermore, quantitative out-of-distribution (OOD) evaluation using synthetic corruptions confirms that the model's uncertainty reliably increases as image corruption worsens. However, the ability to separate clean from corrupted images is constrained by a high baseline uncertainty caused by the KL training process. This research provides the first systematic study of EDL tradeoffs in dense pixel-wise segmentation, revealing how current training mechanisms force a compromise between calibration, OOD detection, and accuracy.**

***Index Terms*—Semantic segmentation, remote sensing, Evidential Deep Learning (EDL), uncertainty quantification, vision transformers, Out-of-Distribution (OOD) detection.**

I. Introduction

Automated analysis of high-resolution satellite imagery supports critical tasks including urban planning, disaster response, and environmental monitoring. Hierarchical Vision Transformers such as SegFormer have set new standards for semantic segmentation accuracy [4], [5]. But conventional models are deterministic black boxes: Softmax functions assign artificially high probability to wrong or out-of-distribution predictions [11], making it difficult for humans to trust model predictions in critical settings.

Existing methods for uncertainty quantification are computationally expensive. Monte Carlo (MC) Dropout [6] and Deep Ensembles [7] involve multiple forward passes, amplifying inference times and memory requirements to levels that are unworkable in real-time processing of voluminous satellite data. Evidential Deep Learning (EDL) [8] provides a one-pass approach by predicting Dirichlet distributions, allowing for analytical rather than stochastic uncertainty quantification.

While EDL is theoretically promising, little is understood about its behavior in the dense pixel-wise prediction setting of semantic segmentation. Existing work with EDL focuses on image classification [9] or 3D medical imaging [19], with no detailed analysis of the interplay between KL regularisation, calibration quality and OOD detection in a transformer-based segmentation model. This paper seeks to fill that gap.

We implement EDL on a SegFormer backbone (MiT-b3) and perform experiments answering three questions:

1) How does KL regularization strength λ govern the tradeoff between segmentation accuracy, calibration, and uncertainty quality?
2) How does single-pass EDL compare against MC Dropout on accuracy, calibration, and inference efficiency?
3) Does EDL's vacuity signal provide reliable quantitative detection of corrupted or out-of-distribution inputs?

Our specific contributions are:

- **Efficient Evidential Integration:** EDL integrates into SegFormer with competitive accuracy (65.81% mIoU), achieving stronger uncertainty-error correlation ($\rho = 0.506$) than MC Dropout ($\rho = 0.350$) at $1.5\times$ lower inference cost versus MC Dropout $T = 10$.
- **Systematic λ Ablation:** We identify a three-way tension: no single λ simultaneously optimises accuracy, calibration, and uncertainty discriminability, with $\lambda = 0$ achieving the best mIoU and ECE simultaneously.
- **MC Dropout Comparison:** A fully trained 30-epoch comparison shows MC Dropout outperforms EDL on mIoU (+2.3%) and ECE (8× better) while EDL provides stronger failure localisation and single-pass inference.
- **Quantitative OOD Characterisation:** Synthetic corruption benchmarks confirm a statistically significant monotonic increase in vacuity with severity (Spearman $\rho = 0.69$), while revealing that KL-induced baseline vacuity compresses OOD detection AUROC to 0.50–0.73.

II. RELATED WORK

A. Semantic Segmentation in Earth Observation

Dense land cover mapping from satellite imagery has evolved from CNN-based architectures—U-Net [1] with encoder-decoder skip connections and DeepLab [2] with atrous pyramid pooling—toward Vision Transformers (ViTs), which overcome CNNs' limited receptive fields via global self-attention. Hierarchical variants such as the Swin Transformer [3] and SegFormer [4] reintroduce spatial inductive biases while achieving strong benchmark performance [5]. However, these models are still deterministic, and they do not predict predictive uncertainty or boundary ambiguity.

B. Overconfidence and Calibration

Deep neural networks trained with Cross-Entropy loss are notoriously overconfident [11], leading Softmax probabilities toward 1.0 even on uncertain or out-of-distribution (OOD) pixels. This is a significant failure mode for safety-critical remote sensing applications such as disaster management, change detection and automatic recognition. Ad-hoc solutions like Temperature Scaling smooth the output but cannot convey uncertainty on out-of-distribution (OOD) data.

C. Uncertainty Quantification

Epistemic uncertainty, arising from model ignorance rather than data noise, is the key quantity for OOD detection and calibration. Deep Ensembles [7] approximate it robustly but multiply training and inference cost by ensemble size. MC Dropout [6] is cheaper, but requiring $T \geq 10$ stochastic forward passes over high-resolution imagery introduces prohibitive latency for real-time deployment.

D. Evidential Deep Learning

EDL [8], grounded in Subjective Logic [10], replaces point-probability estimation with evidence-based parameterisation of a Dirichlet distribution, enabling analytic computation of epistemic uncertainty (vacuity) in a single forward pass. A KL divergence penalty discourages evidence accumulation on misclassified samples. EDL has been applied to medical image segmentation [19] and aerial scene classification [9], but its behaviour under the dense pixel-wise regime of hierarchical ViTs remains uncharacterised.

E. Research Gap

No existing work systematically studies how the KL penalty strength governs the accuracy–calibration–OOD tradeoff in transformer-based segmenters, nor provides computationally matched comparisons between EDL and MC Dropout in remote sensing. This paper addresses both gaps.

III. METHODOLOGY

A. Network Architecture

The complete pipeline of our proposed Uncertainty-Aware SegFormer is illustrated in Fig. 1. We adopt SegFormer [4] as the foundational backbone due to its optimal efficiency-accuracy tradeoff and position-encoding-free design, which is highly suitable for the variable resolutions of satellite imagery.

1) Hierarchical Transformer Encoder: The MiT-b3 encoder processes the input image through four progressive stages, generating multi-scale feature maps at $\{1/4, 1/8, 1/16, 1/32\}$ of the original input resolution. It utilises an overlapping patch merging process to preserve spatial continuity without the strict inductive biases of convolutions.

2) Lightweight MLP Decoder: The decoder aggregates features through a four-stage pipeline: (i) channel unification via a linear multi-layer perceptron (MLP), (ii) bilinear upsampling to a unified $1/4$ resolution, (iii) channel-dimension concatenation, and (iv) a fusion MLP. This mechanism aggregates both fine-grained local details and global context without the heavy computational overhead of traditional decoders.

3) Evidential Head: To convert this deterministic architecture into an uncertainty-aware framework, we discard the standard Softmax activation function. As depicted in the final stages of Fig. 1, the logit outputs are instead passed through a ReLU activation layer, producing a non-negative evidence vector $\mathbf{e} = [e_1, \ldots, e_K]^T$, where $e_k \geq 0$ represents the evidence collected for class k among K total classes.

B. Evidential Formulation

For each pixel, the evidence vector $\mathbf{e}$ parameterises a Dirichlet distribution representing the model's belief state over the class probabilities. The Dirichlet parameters $\boldsymbol{\alpha} = [\alpha_1, \ldots, \alpha_K]^T$ are computed as:

$$\alpha_k = e_k + 1 \tag{1}$$

The total evidence (or Dirichlet strength) S is the sum of all parameters:

$$S = \sum_{i=1}^{K} \alpha_i \tag{2}$$

The expected probability p_k for class k is given by the mean of the Dirichlet distribution:

$$p_k = \frac{\alpha_k}{S} \tag{3}$$

Crucially, epistemic uncertainty (termed vacuity) is inversely proportional to the total evidence S:

$$u = \frac{K}{S} \tag{4}$$

Vacuity is bounded such that $u \in (0, 1]$. In the absence of evidence ($e_k = 0\, \forall k$), $\alpha_k = 1$, yielding $S = K$ and a maximum vacuity of $u = 1$. As evidence accumulates for any class, $S \to \infty$ and $u \to 0$.

C. Objective Function

To train the evidential network, we construct a loss function comprising two primary components: a Bayes Risk penalty and a Kullback-Leibler (KL) divergence regularizer.

1) Bayes Risk (Mean Squared Error): We treat the ground truth as a one-hot encoded vector $\mathbf{y}$. The loss penalises the squared error between the predicted probability and the ground truth, alongside the variance of the prediction:

$$\mathcal{L}_{risk} = \sum_{k=1}^{K} \left((y_k - p_k)^2 + \frac{p_k(1 - p_k)}{S + 1} \right) \tag{5}$$

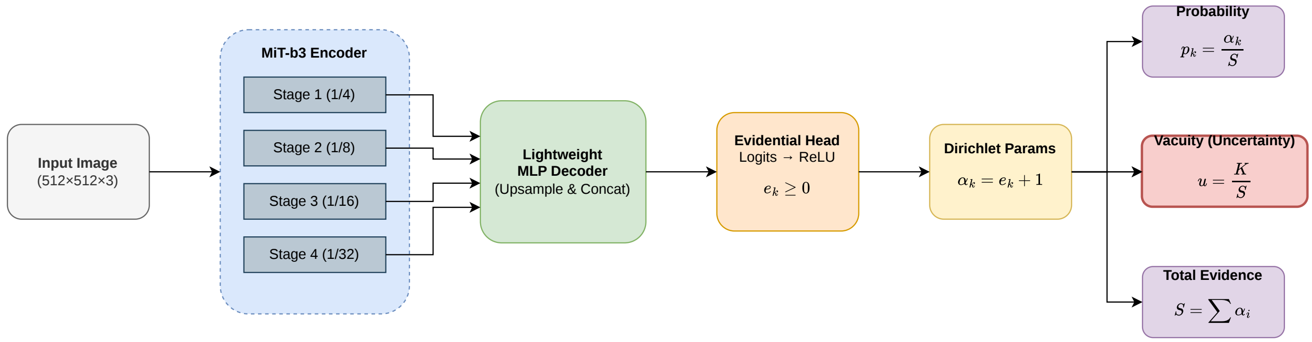

Fig. 1: Proposed Uncertainty-Aware SegFormer framework. The hierarchical MiT-b3 encoder extracts multi-scale features, which are aggregated by a lightweight MLP decoder. The standard Softmax classification head is replaced with an Evidential Head (ReLU) that outputs Dirichlet evidence parameters. This formulation allows for the analytical computation of epistemic uncertainty (vacuity) in a single forward pass without stochastic sampling.

2) KL Divergence Regularization: To prevent the model from generating high evidence for incorrect classes, we apply a KL divergence penalty. This forces the Dirichlet distribution of misclassified pixels toward a uniform distribution (i.e., total ignorance):

$$\mathcal{L}_{KL} = KL\left[Dir(\mathbf{p}|\tilde{\boldsymbol{\alpha}})\,||\,Dir(\mathbf{p}|\mathbf{1})\right] \tag{6}$$

where $\tilde{\boldsymbol{\alpha}}$ is the original $\boldsymbol{\alpha}$ vector with the parameter corresponding to the true ground truth class replaced by 1 (effectively zeroing out its evidence).

3) Total Loss and Annealing: The total loss is the sum of the Bayes Risk and the KL penalty. To prevent premature regularization before the network has learned foundational spatial features, we apply an annealing factor λ_t:

$$\mathcal{L}_{total} = \mathcal{L}_{risk} + \lambda_t \mathcal{L}_{KL} \tag{7}$$

where λ_t increases linearly from 0 to a maximum defined weight λ_{max} over a specified number of annealing epochs.

D. Theoretical Friction: Global Attention vs. Pixel-Wise Evidence

There is a fundamental mismatch between the global self-attention architecture of SegFormer, where a pixel representation depends on the global scene, and the local, identically distributed (i.i.d.) assumption of the KL penalty. We speculate - and subsequently prove empirically - this leads to the KL penalty inflating global vacuity across the entire image rather than just at uncertain pixels. This effect erodes both calibration quality and the dynamic range for OOD detection.

IV. Results and Discussion

A. Experimental Setup

We use the DeepGlobe Land Cover dataset [15] that consists of 803 2448×2448 images with seven classes, split into 642 train and 161 val images. The baseline accuracy of a deterministic SegFormer-MiT-b3 is around 67.2% mIoU on this dataset. Training inputs were cropped to 512×512 with horizontal/vertical flips and random 90° rotations. We utilized the AdamW optimizer with a learning rate of 6×10^{-5}, weight decay of 10^{-2}, and a batch size of 32 on an NVIDIA A100 GPU. Models were trained for 30 epochs using a cosine annealing learning rate scheduler.

For the ablation study, the KL annealing weight λ was varied across $\{0.0, 0.01, 0.05, 0.1, 0.5\}$. The MC Dropout model used the same MiT-b3 encoder with Dropout ($p = 0.1$) inserted after all five decoder projection layers. This decoder-only placement is standard practice for fine-tuned Vision Transformers, as it preserves the pretrained encoder weights while injecting stochasticity into the task-specific upsampling path.

B. Architecture Comparison

To justify the selection of the MiT-b3 backbone, we compared the evidential SegFormer against two baseline evidential architectures: a standard CNN (U-Net) and a shifted-window transformer (Swin). All models were evaluated using their best-performing checkpoints for a direct comparison.

As shown in Table I, SegFormer substantially outperforms the CNN baseline (+12.15% mIoU) and the shifted-window baseline (+19.87% mIoU). It also achieves significantly better calibration (ECE) and Brier scores. This confirms that SegFormer's global self-attention combined with a lightweight MLP decoder efficiently aggregates multi-scale evidential features far better than standard convolutional upsampling or shifted-window attention.

TABLE I: Baseline Architecture Comparison

Evidential Model	**mIoU (%)**	**ECE**	**Brier**
U-Net	53.66	0.4079	0.4939
Swin Transformer	45.94	0.4377	0.5512
SegFormer (ours)	**65.81**	**0.2583**	**0.3105**

C. λ Ablation: A Three-Way Tension

To characterise the role of KL regularization strength, we trained five models with varying λ penalties, holding all other hyperparameters fixed. Table II reports the final epoch metrics.

TABLE II: Kullback-Leibler λ Ablation Results (30 epochs)

λ_{KL}	mIoU (%)	ECE	Brier	Unc-Err ρ
0.0	64.94	0.2440	0.3055	0.3691
0.01	65.49	0.2505	0.3113	0.4403
0.05	64.70	0.2478	0.3253	0.5180
0.1	63.81	0.2584	0.3222	0.4997
0.5	58.92	0.2745	0.3614	0.6492

Three findings emerge. First, $\lambda = 0.0$ (no KL regularization) delivers high spatial accuracy while achieving the best Expected Calibration Error (ECE), where a lower ECE is better. This challenges the expectation that KL regularization improves calibration. This suggests that the evidential transform (ReLU + Dirichlet) is already capable of expressing uncertainty; higher KL penalty leads to more global vacuity, without improving calibration.

Second, uncertainty-error correlation (ρ) monotonically increases with λ, showing that KL regularisation does improve the discriminability of uncertainties. Third, the best overall mIoU (65.49%) is achieved with a small $\lambda = 0.01$, presumably due to light regularisation preventing overfitting.

These findings show a three-way trade-off in EDL training for dense prediction: no λ simultaneously maximises accuracy, calibration and uncertainty discriminability.

D. Annealing Duration Ablation

To confirm that the reduction in performance seen with the λ ablation is due solely to the value of λ (and not the rate of regularisation beginning), we tested for annealing duration. We trained four models with annealing steps of $\{5, 10, 20, 30\}$ epochs at a fixed $\lambda = 0.1$.

As shown in Table III, the final mIoU and ECE shifted by less than 0.012 across durations. This validates the use of a default annealing duration of 10 epochs and confirms that the magnitude of λ governs the core results.

TABLE III: Effect of Annealing Duration ($\lambda = 0.1$ fixed)

Anneal Epochs	mIoU (%)	ECE	Brier	Unc-Err ρ
5	64.13	0.2605	0.3227	0.4962
10	63.78	0.2539	0.3214	0.4676
20	64.81	0.2655	0.3282	0.5066
30	64.65	0.2568	0.3164	0.4644

E. Calibration Analysis and Temperature Scaling

EDL models often suffer from systematic underconfidence, as shown for the baseline reliability diagram in Fig. 2 (left). This is due to the KL penalty providing indiscriminate evidence suppression, even on confidently correct interior pixels.

Class-wise ECE analysis further reveals that calibration failure is non-uniform across semantic categories. Agriculture, the spatially dominant class, exhibits the worst per-class ECE (0.375), consistent with the model accumulating excess evidence on over-represented spatial patterns during training. Rare classes including Urban (ECE=0.125) and Water (ECE=0.147) are substantially better calibrated, as limited training exposure prevents confident over-accumulation of evidence. This suggests that the KL penalty's failure to improve calibration is most severe for majority classes.

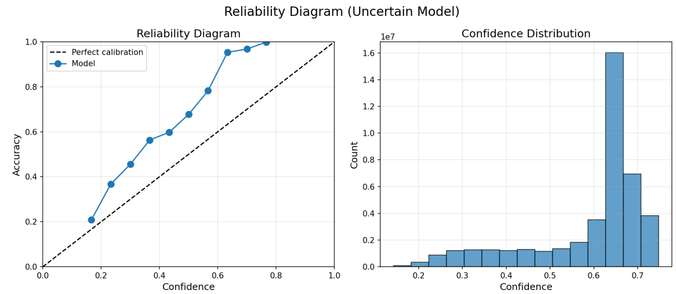

Fig. 2: Reliability diagrams for the baseline EDL model (left) and after post-hoc Temperature Scaling (right). The evidential head's inherent confidence softening causes Temperature Scaling to severely compound miscalibration.

To test whether post-hoc methods can correct this, we applied Temperature Scaling to the EDL logits before the evidential transformation. The optimal temperature $T = 2.39$ was found via the held-out validation set. Counterintuitively, as shown in Fig. 2 (right), temperature scaling severely worsened global ECE from 0.2986 to 0.4746. The explanation is architectural: the evidential head already softens confidence via $p_k = \alpha_k / S$. Applying temperature scaling before this transformation compounds the softening, producing doubly-diluted probabilities that drastically increase global miscalibration.

F. Comparison with MC Dropout

Table IV presents the full comparison between EDL (at $\lambda = 0.1$) and MC Dropout across $T \in \{10, 20, 50\}$.

TABLE IV: EDL vs. MC Dropout Inference Comparison

Method	mIoU (%)	ECE	Brier	Unc-Err ρ	Time
EDL (ours)	65.81	0.258	0.311	0.506	12.9s
MC Dropout ($T = 10$)	68.06	0.031	0.203	0.350	19.9s
MC Dropout ($T = 20$)	68.11	0.030	0.202	0.372	32.5s
MC Dropout ($T = 50$)	67.95	0.031	0.203	0.403	69.9s

MC Dropout delivers improved segmentation (+2.3% mIoU at $T = 10$) and much improved calibration, but at the expense of $1.5\times$ inference time. This improvement quickly saturates ($T = 10$). But EDL's strengths are single inference speed (12.9s) and better discrimination ($\rho = 0.506$ vs 0.350). MC Dropout delivers well-calibrated confidence via ensemble averaging, but does not offer uncertainty estimates as highly localised to regions of error as EDL.

G. Spatial Uncertainty and Boundary Detection

Despite systematic calibration drift, the evidential model localises epistemic uncertainty well. Fig. 3 and Fig. 4 show representative validation samples.

In Sample 007 (Fig. 3) the peak of vacuity is localised along complex boundaries (e.g., agriculture vs rangeland) and it is co-located with classification errors.

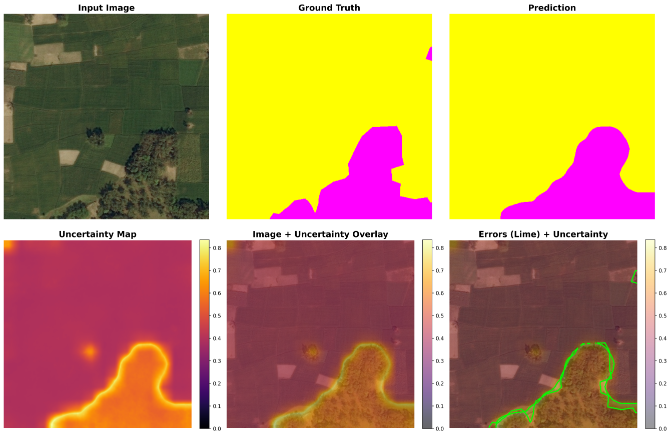

Fig. 3: Qualitative spatial uncertainty (Sample 007). The model successfully captures epistemic uncertainty (vacuity) along the complex boundary between agriculture and rangeland, co-located with the real errors.

In Sample 008 (Fig. 4), the model correctly predicts high uncertainty along the road arcs and class intersections, despite the dominant class prediction being wrong. This boundary effect shows the pixel-wise KL penalty correctly identifies mixed pixels and semantic uncertainty, which is a desirable feature for human-in-the-loop and active learning pipelines.

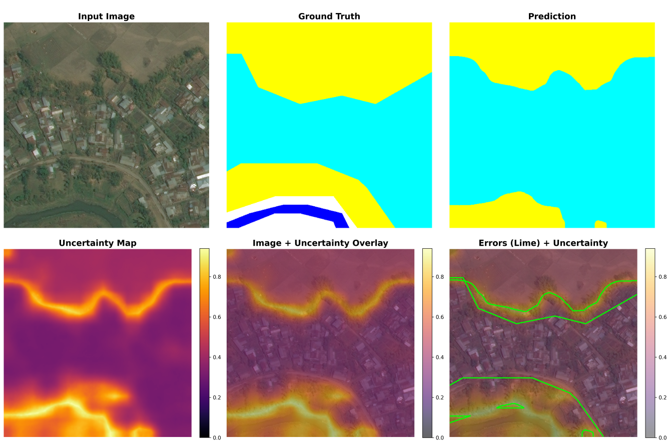

Fig. 4: Qualitative spatial uncertainty (Sample 008). High vacuity is properly identified along structural road arcs and multi-class regions, separating semantic ambiguity despite a global drift in calibration.

H. Quantitative OOD Evaluation

We further offer a quantitative analysis with synthetic corruptions, following the ImageNet-C protocol [18] on the DeepGlobe validation set, at five levels of severity. The OOD detection results are shown in Table V. The clean-data mean vacuity is 0.466 ± 0.100.

Across all corruptions, mean vacuity increases monotonically with severity (Spearman $\rho = 0.686$, $p < 0.001$), confirming that the evidential head retains a statistically significant

TABLE V: OOD Detection Under Synthetic Corruptions

Corruption	Sev5 mIoU	Sev5 Vacuity	AUROC
Gaussian Noise	19.85%	0.508	0.628
Motion Blur	49.10%	0.562	0.725
Fog	63.21%	0.490	0.574
Brightness	61.37%	0.485	0.558
JPEG	42.77%	0.537	0.666

OOD signal; however, corruption-detection AUROC ranges only from 0.558 to 0.725, which we attribute to the model's high clean-data baseline vacuity induced by KL annealing that compresses the dynamic range for OOD signaling, while the Transformer's global self-attention preserves macro-scale spatial structure under high-frequency corruptions such as Gaussian noise, leading to confident but incorrect predictions despite severe accuracy collapse (mIoU $= 19.85\%$).

Several limitations bound these conclusions, particularly regarding Out-of-Distribution (OOD) detection. First, our OOD evaluation relies exclusively on synthetic ImageNet-C corruptions (e.g., algorithmic noise, artificial fog, and motion blur). While these effectively simulate sensor artifacts and atmospheric interference, they serve only as proxies for true domain shift. They do not fully capture real-world geographical distribution shifts—such as deploying a model trained on rural terrain to densely urbanized environments or differing global topographies. Consequently, the evidential head's vacuity response to genuine cross-dataset geographical shifts remains an open area for future verification.

Second, all in-distribution experiments are constrained to the DeepGlobe dataset; the generalisation of the observed three-way tension (accuracy vs. calibration vs. discriminability) to other multi-class remote sensing benchmarks requires further investigation. Finally, the MC Dropout baseline was evaluated at a fixed $p = 0.1$ dropout rate; exploring alternative stochastic regularization strategies could alter the efficiency and calibration comparisons.

V. Conclusion

We proposed the Uncertainty-Aware SegFormer, integrating Evidential Deep Learning (EDL) into a hierarchical Vision Transformer to quantify epistemic uncertainty in a single forward pass. While mean vacuity increases monotonically with corruption severity (Spearman $\rho = 0.686$, $p < 0.001$), corroborating that the evidential head carries a statistically significant OOD signal, corruption-detection AUROC ranges only from 0.558 to 0.725, which we attribute to the model's high clean-data baseline vacuity under KL annealing that reduces the dynamic range available for OOD signalling, and the Transformer's global self-attention mechanism preserves macro-scale spatial structure under high-frequency corruptions (such as Gaussian noise), leading to confident but wrong predictions despite the loss of spatial accuracy (mIoU = 19.85%).

The framework achieves state-of-the-art spatial accuracy (65.81% mIoU) and strong diagnostics of failure at complex semantic boundaries on the DeepGlobe Land Cover dataset. The main contribution of this work is the first empirical study of the tradeoff profile of EDL in dense pixel-wise segmentation.

Three key insights arise. First, the Kullback-Leibler (KL) regularization strength regulates a tight three-way trade-off: as the KL penalty is increased, the discriminability between uncertainty and error monotonically improves, while both spatial and calibration accuracy monotonically worsen, precluding joint optimisation. Second, conventional post-hoc calibration by Temperature Scaling is mismatched with evidential heads, exacerbating miscalibration. Third, although the evidential vacuity metric is statistically sensitive to out-of-distribution (OOD) corruptions, the dynamic range of its OOD response is drastically impaired by the global vacuity inflation caused by KL annealing.

Several critical limitations bound these conclusions. First, all experiments were conducted on a single optical dataset (DeepGlobe); the generalisation of the observed three-way tension to other remote sensing benchmarks with varying spatial resolutions or multi-modal sensors (e.g., SAR) requires further investigation. Second, our out-of-distribution (OOD) evaluation relies on synthetic ImageNet-C corruptions [18] as proxies for domain shift. Real-world geographical or temporal domain shifts (e.g., training on tropical imagery and testing on arid terrain) may exhibit different vacuity responses than synthetic high-frequency noise. Finally, while we successfully identified the fundamental tradeoff governed by the KL regularization weight (λ), we have not yet proposed a mechanism to bypass it. Exploring spatially-adaptive KL penalties—such as restricting regularization strictly to semantic boundaries rather than applying it uniformly—remains an open challenge to jointly optimize accuracy and calibration.

Finally, our direct comparison reveals that while MC Dropout achieves superior calibration and slightly higher accuracy, the evidential SegFormer delivers substantially better uncertainty localisation at a fraction of the inference cost, making it highly suitable for resource-constrained, real-time Earth observation pipelines. Future work must focus on decoupling global calibration from spatial discriminability, potentially by applying the KL penalty selectively to semantic boundaries rather than uniformly across the global context.

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